

Crack Detection Using Multiple Image Processing for Unmanned Aerial Monitoring of Concrete Structure

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Abstract— Concrete structures are subjected to crack formation over time. The causes of crack formation are shrinkage, thermal stress, weathering, etc. This paper is about the construction and application of a device that uses image processing to detect cracks. The system is carried by a drone and can be remotely accessed. The system has a graphical user interface for initializing the device, viewing real time video, taking pictures of a crack, measuring its width, and evaluating if safe or unsafe. The pictures and their results are stored for reference.

Keywords— Concrete crack, image processing, drone

I. INTRODUCTION

Concrete buildings no matter how big or strong deteriorate over time. Buildings weaken with the presence of cracks. A crack is defined as the splitting of an object or a certain material into two, or more under the action of stress. Depending on the cracked substances, these cracks can reduce the strength of the materials in most cases and sometimes, it degrades their quality [1]. These inevitable loadings or calamities may cause the structural deterioration and damage that would risk to a higher chance of catastrophic collapses. Thus, a wide variety of maintenance strategies for the structural safety are developed. Crack information represents the current structural health of a certain civil infrastructures [2]. These cracks form due to various reasons such as the permeability of concrete namely – water-cement ratio, degree of hydration/curing, etc.; and thermal movement or expansion and contraction due to heat [3].

In traditional method, the way on determining the defects and cracks on concrete surfaces are done manually by inspection. Additionally, any internal and hazardous defects on the concrete would require destructive method for detection [4]. The problem lies in the difficulty on determining the cracks in the higher part of the structure that is unreachable by manual inspection methods.

The goal of this study is: (1) to apply the sequential image processing algorithms for crack detection and measurement; (2) to create the hardware of the system as a payload of an unmanned aerial vehicle; and (3) to implement and test the prototype in actual buildings made of concrete structures, with the aid of piloting the drone and to determine whether the detected crack is safe or unsafe.

The study would be beneficial to the engineers for the inspection of the concrete structure for the crack formation from unreachable place.

The scope of this project is to use suitable algorithms from the existing image processing techniques for the wall crack detection system. The researchers used a Raspberry Pi 3 model B with a camera and ultrasonic sensor, assembled to be carried by a drone. The crack detection was processed by the Raspberry Pi 3 model B. The main structural target for this research is concrete building and any other higher structures where crack locations are not easily reached, and the cracks should be visible. The device can only measure cracks of certain length and it can only measure them at a maximum distance. Only the coverage of the camera for the crack length will be processed since the system did not implement image stitching.

II. REVIEW OF RELATED LITERATURE

Cracks are often categorized as occurring either in plastic or in hardened concrete. The cracks in the hardened concrete is the main focus of this research. Thus, the causes of cracks in the hardened concrete are Drying Shrinkage, Thermal Stresses, Chemical Reaction, Weathering, Corrosion of reinforcement, Poor Construction Practices, Construction Overloads, Errors in Design and Detailing, and Externally Applied Loads [4].

From all these causes, according to Grishna Thagunna, thermal movement is a more serious threat to concrete because it is exposed to heating and cooling, forcing it to expand or to contract [5]. Moisture movement is a common cause of crack formation since this will cause the cement paste to lose its moisture, hence causing it to shrink. Absorbing moisture will expand the material. The moisture induced volume change is an inherent characteristic of the concrete.

From the study conducted by the American Concrete Institute (ACI), there is a limit for tolerating crack widths of reinforced concretes under service loads. Concrete buildings that are exposed to different environments such as dry air, chemicals, or humidity has different tolerance for crack widths. Table 1 shows the general guide for proportioning reinforcement in design [4].

TABLE I. GUIDE TO REASONABLE CRACK WIDTHS

Exposure Condition	Crack Width	
Dry air or protective membrane	0.016 in.	0.41 mm
Humidity, moist air, soil	0.012 in.	0.30 mm
Deicing chemical	0.007 in.	0.18 mm
Seawater, wetting and drying	0.006 in.	0.15 mm
Water-retaining structure	0.004 in.	0.10 mm

It is said that the grayscale is the foundation for subsequent image processing. The grayscale is used for peak measurement, mean value, and weighted value [6].

Thresholding is one of the classifications of Image Segmentation and it is the simplest method in segmentation. Threshold method is usually used in binary image conversion [7].

Morphological operations are methods for processing binary images and for gray scale images based on shapes. Defects are extracted from the background by thresholding the image and classified according to size and shape parameters [8].

Integrated image processing algorithms are used for pattern recognition in identifying cracks and defects from concrete surfaces. The image processing used includes Hough transform, LoG weighted, Dilation, Grayscale, Canny Edge Detection and Haar Wavelet Transform [9].

Raspberry Pi was used in creating an environmental wireless sensor network for greenhouse monitoring of gases. The built in Wi Fi of the Raspberry Pi was used to establish the wireless network between the nodes and the data aggregator in the integration of environmental sensors [10]. This proves the capability of Raspberry Pi's Wi-Fi to handle data with an increase in transfer rate, compared to other technologies such as Bluetooth.

III. METHODOLOGY

Figure 1 shows the framework of the system. The input is the surface image of the concrete and the distance of the wall to the device. The process in sequential order is: grayscale, adaptive threshold, invert, median filter, image morphology, fitted contour, and skeletal morph. It was done by the Raspberry Pi 3 model B. After processing, the output was displayed on the laptop, which contains the crack width, crack length, distance, and crack condition.

Figure 2 shows the process in the Raspberry Pi for crack detection and measurements. As observed from the figure, the device will take a picture and will be converted to grayscale format by using Equation 1. R' is the red layer intensity, G' is the green layer intensity, B' is the blue layer intensity.

$$Im[x,y] = 0.2989 R' + 0.5870 G' + 0.1140 B' \quad (1)$$

After converting to grayscale, the adaptive threshold was used to extract the desired region, referring to the crack, using the Equations 2 and 3. $I[x,y]$ is the pixel value, 1 is white, and 0 is black, $Im(x,y)$ is the intensity value and T is the threshold value. $WA(x,y)$ is the weighted average of an $N \times N$ matrix.

$$I[x,y] = \begin{cases} 1, & Im[x,y] > T \\ 0, & Im[x,y] \leq T \end{cases} \quad (2)$$

$$T = WA(x,y) \quad (3)$$

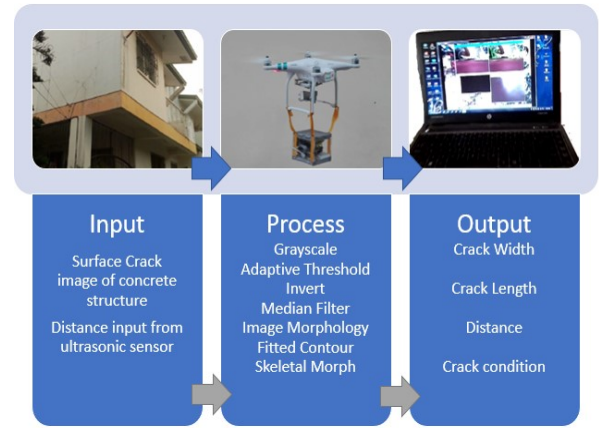


Fig. 1. Framework of crack detection using image process with drone.

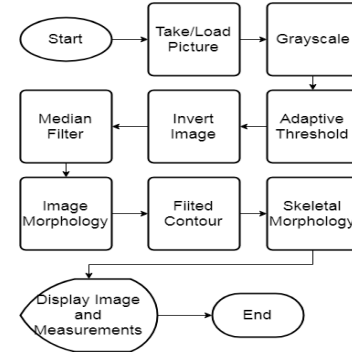


Fig. 2. Crack detection and measurement in Raspberry Pi 3 model B.

The image was then inverted for pre-processing before median filter and image morphology. After inverting, a median filter is used for denoising and is done using Equation 4. The $Y[m,n]$ is the median value from the matrix, $[m,n]$ is the center of the matrix. The $x[i,j]$ is the intensities values, $[i,j]$ is the location of the $x[i,j]$ and w is the pixels in the neighbourhood.

$$y[m,n] = \text{median} \{x[i,j], (i,j)w\} \quad (4)$$

The excess blob which has an area less than 50 was removed by image morphology using Equation 5. A is original image, B is region with less than 50 pixels.

$$I = (A \ominus B) \oplus B \quad (5)$$

A fitted contour, rectangular in shape, was used to locate the crack region in the image and using the twice of the major axis of the rectangle as the crack length in pixels. For the crack width in pixel extraction, a skeletal morphology was applied by using Equation 6. A is the image, K is the value of the iterative such that A erodes into an empty set. S_k is the skeletal subset. The 'k' is successive erosion of A by B . After applying it, calculating the nearest distance between skeletal pixels provides the width in pixels

$$A = \bigcup_k^K (S_k(A) \oplus kB) \quad (6)$$

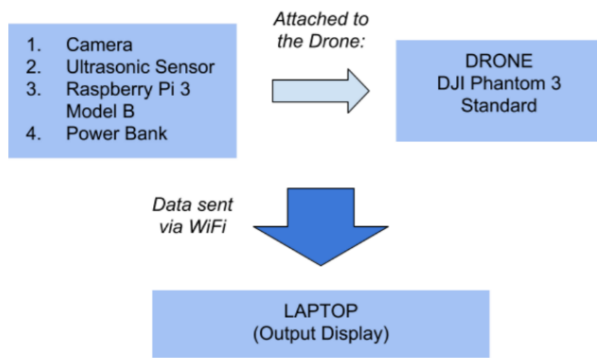


Fig. 3. Diagram of crack detection system.

The diagram shown in the Figure 3 is about the utilization of the function of the raspberry pi model 3. The pi camera and the ultrasonic sensor are connected to the raspberry pi 3 and will be supplied by the power bank and it will be the payload for the drone. The payload was lifted such that it will be able to monitor the surface of the structure for crack formation using the pi camera and measure the distance from the payload to the surface. The data gathered and processed by the raspberry pi is then sent to the Laptop, via Wi-Fi. This raspberry pi evaluates the processed image of cracks whether the cracks are still safe or not. The Laptop displays the results, showing the image of the crack; safeness of the crack and its corresponding width, crack length and the distance of the drone from the crack.

Before the actual implementation of the said research, it is essential to test the built prototype in detecting actual cracks. In this time, the drone, with the device attached, was controlled to fly at a low altitude. This was done to create a more accurate calibration because flight is a lot stable at low height and the actual crack width and distance can be measured easily. Actual implementation will start if the prototype functions correctly. This means that the camera will capture the crack image and the algorithm will do the calculations. It is possible to have some certain errors if the image has some noise integrated instead of the crack. This means that the image noise will possibly be calculated even though it is not an actual crack.

A single crack from a concrete wall was chosen and the device was tested by taking pictures of the same crack from different distances. This means that there is a predetermined crack width and the resulting crack width from varying distances should be the same. A total of 10 tests were done. To calculate the real crack width, the system was then calibrated by averaging the calculated scaling coefficients.

Equations 7 and 8 are used to determine the scaling coefficient. These equations were used to calculate the actual crack width and length [2]. W_R is the real crack width, α is the scaling coefficient, W_P is the crack width in pixels, and D is the distance between the camera and the cracks. L_R is the actual crack length, and L_P is the length in pixels.

$$W_R = \alpha W_P D \quad (7)$$

$$L_R = \alpha L_P D \quad (8)$$

TABLE II. GATHERED VALUES DURING CALIBRATION

Trial	Actual Crack Width (mm) W_R	Crack Width in Pixels W_P	Distance (cm) D	Scaling Coefficient α
1	1.016	4.919	10.729	0.019251
2	1.016	4.714	10.988	0.019615
3	1.016	2.107	23.492	0.020526
4	1.016	1.978	26.001	0.019755
5	1.016	1.545	33.438	0.019666
6	1.016	1.406	36.377	0.019865
7	1.016	1.351	37.917	0.019834
8	1.016	1.224	40.235	0.020630
9	1.016	1.142	46.311	0.019211
10	1.016	0.897	57.556	0.019679
Average Scaling Coefficient				0.019803
Standard Deviation				0.000464

As shown in Table 2, ten trials were done wherein each consecutive trial has an increase in distance between the device and the crack. Equation 1 is used on getting the scaling coefficient. Upon observation, the computed scaling coefficient of each trial has insignificant differences. These computed values were added together and divided by the number of trials to compute for the average. The resulting average of 0.019803 ± 0.000464 was the scaling coefficient used in the program. The scaling coefficient was then applied to Equations 7 and 8. To evaluate whether the crack is safe or not, the calculated width was compared to a standard threshold value of 0.41mm [4]. If the measured value is less than 0.41mm, then it is safe, otherwise unsafe.

IV. RESULTS AND DISCUSSION

In testing the developed program, the researchers had chosen a single building containing visible cracks, as shown Figures 4 and 5, for the correction and calibration of the device's crack detection capability.

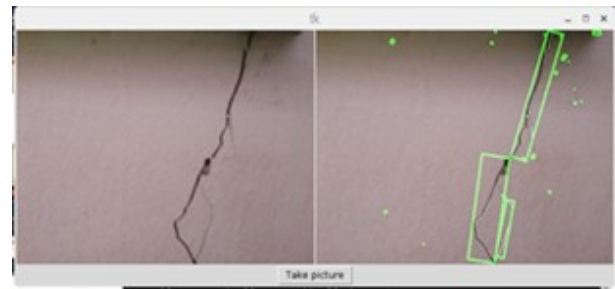


Fig. 4. Concrete surface with crack.



Fig. 5. Concrete surface without crack.

The first evaluation done was testing the crack detection capability of the device as shown in Table 3. The device was tested with 10 pictures of surfaces like in Figure 4 with actual crack, and another 10 pictures of surfaces without cracks like in Figure 5. In this part, confusion matrix was used for the visualization of the performance of an algorithm. The proposed method for crack detection has resulted to have a misclassification rate of 10% and False positive rate of 20%. The device shadows and chapped paint of the surface concrete also contribute to the error.

Based from the result from a picture taken on an actual crack, the crack detection. The sequence of algorithm works and was able to detect cracks. The adaptive thresholding was able to convert the grayscale to binary image and it works well since the crack region was identified even though other noises were present. The median filter was able to remove the “salt and pepper” from the binary picture and leaving only the region with white pixels. Those noisy areas were removed by the image morphology wherein area with less than 50 pixels will be removed. The fitted rectangle is superimposed on the original picture; hence, the crack is detected.

TABLE III. CONFUSION MATRIX

N=20	Predicted No	Predicted Yes
Actual No	8 (TN)	2 (FP)
Actual Yes	0 (FN)	10 (TP)

TABLE IV. CRACK WIDTH MEASUREMENT RESULT FOR BUILDING 1

Crack	Distance (cm)	Actual Width (mm)	Measured (mm)	Percent Error (%)
crack 1	23.9458	1.2700	1.3011	2.4488
	40.0651	1.2700	1.3239	4.2441
	56.6139	1.2700	1.3674	7.6693
crack 2	27.9706	1.0160	1.0012	1.4567
	38.8288	1.0160	1.0842	6.7126
	61.3223	1.0160	1.1157	9.8130
crack 3	40.6609	1.2700	1.2376	2.5512
	48.6820	1.2700	1.3494	6.2520
	64.6159	1.2700	1.3721	8.0394
crack 4	31.3184	1.5240	1.4922	2.0866
	46.4284	1.5240	1.5925	4.4948
	59.2322	1.5240	1.6239	6.5551
crack 5	36.1416	1.2700	1.3399	5.5039
	52.5730	1.2700	1.3473	6.0866
	64.1527	1.2700	1.3733	8.1339
crack 6	18.2431	0.3810	0.3750	1.5748
	29.6121	0.3810	0.4009	5.2231
	49.8087	0.3810	0.4079	7.0604
crack 7	25.6238	0.3810	0.3950	3.6745
	41.2323	0.3810	0.4012	5.3018
	63.4213	0.3810	0.4171	9.4751
crack 8	20.8451	1.2700	1.2532	1.3228
	27.8084	1.2700	1.3421	5.6772
	50.7855	1.2700	1.3799	8.6535
crack 9	19.8481	0.7620	0.7462	2.0735
	30.2321	0.7620	0.7901	3.6877
	62.6531	0.7620	0.8312	9.0814
crack 10	49.9344	1.7780	1.7524	1.4398
	56.1946	1.7780	1.8490	3.9933
	64.3988	1.7780	1.9031	7.0360

The crack samples from Tables 4 and 5 are the same. From these tables, the crack was measured from 3 various distance such that the percent error would not exceed 10 %. As shown in the data, increasing the distance of the device from the surface of the wall would result into a higher error. Table 5 shows the crack length measurement from 3 various distances of the cracks measured on Table 4. It follows a similar trend such that the further the device is, the higher the percent error.

The distance is a limiting factor to the reliability of the system as observed from the data on Tables 4 and 5. The resolution of the crack lowers as the camera distance goes further away from the surface; hence the conversion process is affected and will contribute as an error. Crack widths which is equal or less than 1.778 mm, would have an effective range that is less than 58 cm. At that range, it will provide results with errors that is less than 10 %. The effective range of the system would increase if the crack width is thicker than 1.778 mm; however, the ultrasonic sensor has a max range of 400 cm. At around 40 cm and below, there is a significant decrease in error, only amounting to 5 % or less error.

TABLE V. CRACK LENGTH MEASUREMENT RESULTS FOR BUILDING 1

Crack	Actual Length (mm)	Measured (mm)	Percent Error (%)
crack 1	118.0000	114.9771	2.5618
	118.0000	110.9921	5.9389
	118.0000	108.3215	8.2021
crack 2	234.0000	231.8889	0.9022
	234.0000	229.0857	2.1001
	234.0000	217.4236	7.0839
crack 3	323.0000	315.3321	2.3740
	323.0000	309.3269	4.2332
	323.0000	303.2135	6.1259
crack 4	168.0000	157.1406	6.4639
	168.0000	154.2395	8.1908
	168.0000	151.7658	9.6632
crack 5	228.0000	219.3421	3.7973
	228.0000	216.3231	5.1214
	228.0000	209.2156	8.2388
crack 6	154.0000	149.3327	3.0307
	154.0000	145.7382	5.3648
	154.0000	140.4312	8.8109
crack 7	208.0000	200.4257	3.6415
	208.0000	195.8793	5.8273
	208.0000	190.6463	8.3431
crack 8	36.0000	35.2135	2.1847
	36.0000	33.8934	5.8517
	36.0000	32.6124	9.4100
crack 9	24.0000	23.1235	3.6521
	24.0000	22.9123	4.5321
	24.0000	21.9912	8.3700
crack 10	728.0000	719.2315	1.2045
	728.0000	702.9683	3.4384
	728.0000	691.0458	5.0761

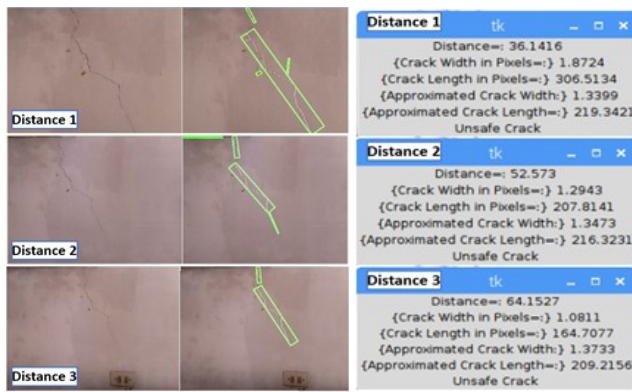


Fig. 6. Processed image of crack at different distances.

In Figure 6, at left side, the picture was taken from three various distances and its corresponding measurements is shown at the right side. It is the data of crack 5 from Table 4. It shows the output of the Raspberry Pi which includes the distances, crack width and length in pixels, measured crack width and length, and its condition.

Table 6 shows the evaluation of the crack samples from Tables 4 and 5. The system was able to evaluate the crack condition by comparing the crack width to the threshold value which is stated in the last part of the methodology. From the results, all crack widths less than 0.41mm were considered safe, otherwise unsafe.

V. CONCLUSION

The study was able to develop a working device for crack detection and measurement by using multiple image processing algorithms. It was able to detect and to measure the crack width and length in pixels. The hardware was constructed in such a way that it can be carried by the drone. The components were also weighted, and the materials used in the assembly of the device were carefully chosen to meet the required payload that the drone to be used can carry (less than 1 kilogram). The sensors used by the Raspberry Pi were calibrated to convert pixel measurement to actual measurement and to provide accurate results. The overall system is capable of detecting cracks as well as measuring the actual crack width and length, then evaluating it whether it is safe or unsafe. The effective range of the system, to provide a percent error less than 10%, is 58 cm or closer from the concrete surface. This error can be minimized by decreasing the distance between the crack and the device. The results and images were saved within the device.

VI. RECOMMENDATION

To further improve this project, the researchers present the following recommendations: (1) Include an algorithm that can determine the crack depth. (2) Improve selectivity or crack detection at dark environment. (3) Include a function to record the location of the crack at the building. (4) Improve wireless access by using better means other than Wi-Fi. (5) Use lighter materials for lesser payload. (6) Higher Resolution Camera, to provide better accuracy. Lastly, (7) Apply image stitch for better coverage on measuring crack length.

TABLE VI. CRACK CONDITION FOR BUILDING 1

CRACK	Measured (mm)	Remarks
crack 1	1.3011	Unsafe
	1.3239	Unsafe
	1.3674	Unsafe
crack 2	1.0012	Unsafe
	1.0842	Unsafe
	1.1157	Unsafe
crack 3	1.2376	Unsafe
	1.3494	Unsafe
	1.3721	Unsafe
crack 4	1.4922	Unsafe
	1.5925	Unsafe
	1.6239	Unsafe
crack 5	1.3399	Unsafe
	1.3473	Unsafe
	1.3733	Unsafe
crack 6	0.3750	Safe
	0.4009	Safe
	0.4079	Safe
crack 7	0.3950	Safe
	0.4012	Safe
	0.4171	Unsafe
crack 8	1.2532	Unsafe
	1.3421	Unsafe
	1.3799	Unsafe
crack 9	0.7462	Unsafe
	0.7901	Unsafe
	0.8312	Unsafe
crack 10	1.7524	Unsafe
	1.8490	Unsafe
	1.9031	Unsafe

REFERENCES

- [1] T. Rashmi R.A; Ashwini D. Mane; Sunil L, "Concrete Structure Analysis and Classification using Image Processing," International Journal of Advanced Computer Research, vol. 3, no. 11, 2013.
- [2] H. K. S. H. S. S. Cho, "Unmanned Aerial Vehicle (UAV)-powered Concrete Crack Detection based on Digital Image Processing," International Workshop on Advanced Smart Materials and Smart Structures Technology, August 1-2, 2015 2015.
- [3] C. Meyer. (2016). Concrete.
- [4] R. J. W. Frosch, Jeffrey S., "Causes, Evaluation, and Repair of Cracks in Concrete Structures," American Concrete Institute, 2007.
- [5] G. Thagunna, "Building Cracks - Causes and Remedies," International Journal of Advanced Structures and Geotechnical Engineering, vol. 4, no. 1, 2015.
- [6] L. Jingyi, W. Ning, L. Ying, and Y. Yuemei, "A Study of Crack Detection Algorithm," in 2015 Fifth International Conference on Instrumentation and Measurement, Computer, Communication and Control (IMCCC), 2015, pp. 1184-1187.
- [7] M. Qiao, W. Xiaoying, and Z. Yu-an, "Research on a least squares thresholding algorithm for pavement crack detection," in 2016 Sixth International Conference on Information Science and Technology (ICIST), 2016, pp. 465-469.
- [8] R. M. C. L. Chandrakar, "Surface Defects Detection for Ceramic Tiles Using Image Processing and Morphological Techniques," IJDW, vol. 4, no. 1, pp. 51-58, 2012.
- [9] J. R. Balbin, *et al.*, "Pattern Recognition of Concrete Surface Cracks and Defects using Integrated Image Processing Algorithms," Second International Workshop on Pattern Recognition, vol. 10443, June 19, 2017 2017. SPIE Proceedings
- [10] R. B. Anire, F. R. G. Cruz, and I. C. Agulto, "Environmental wireless sensor network using raspberry Pi 3 for greenhouse monitoring system," in IEEE 9th HNICE, 2017, pp. 1-5